

CS & IT ENGINEERING



Computer Network

IPv4 Address

Lecture No. - 08

By - Abhishek Sir





Recap of Previous Lecture



Topic

VLSM





Topics to be Covered



Topic

Routing Table

Topic

CIDR

ABOUT ME



Hello, I'm **Abhishek**

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- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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#Q. A company has a class C network address of "204 . 204 . 204 . 0" . It wishes to have three subnets, one with 100 hosts and two with 50 hosts each. Which one of the following options represents a feasible set of subnet address/subnet mask pairs?

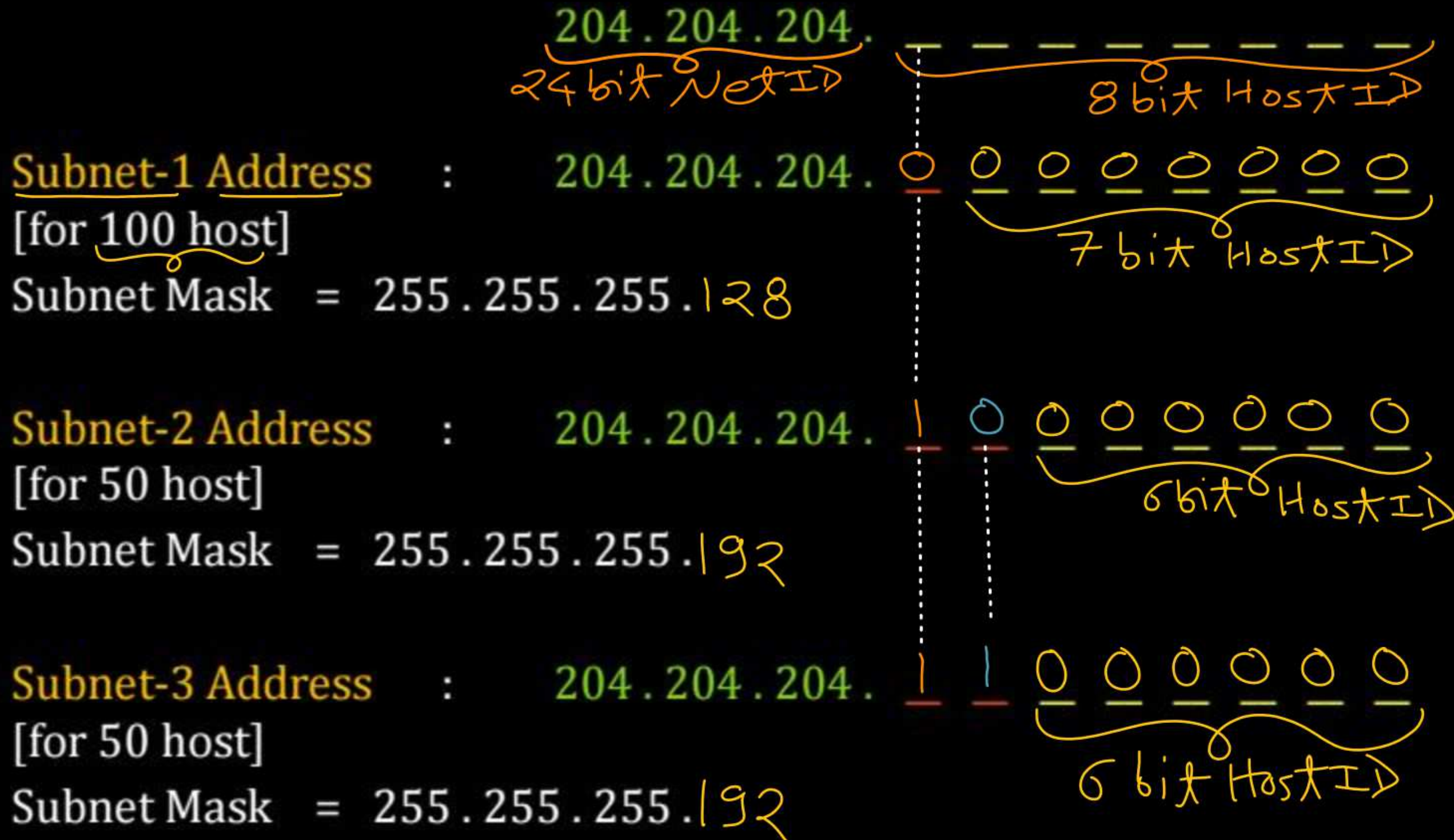
[GATE-IT-2005, 2 Marks]

[MCQ]

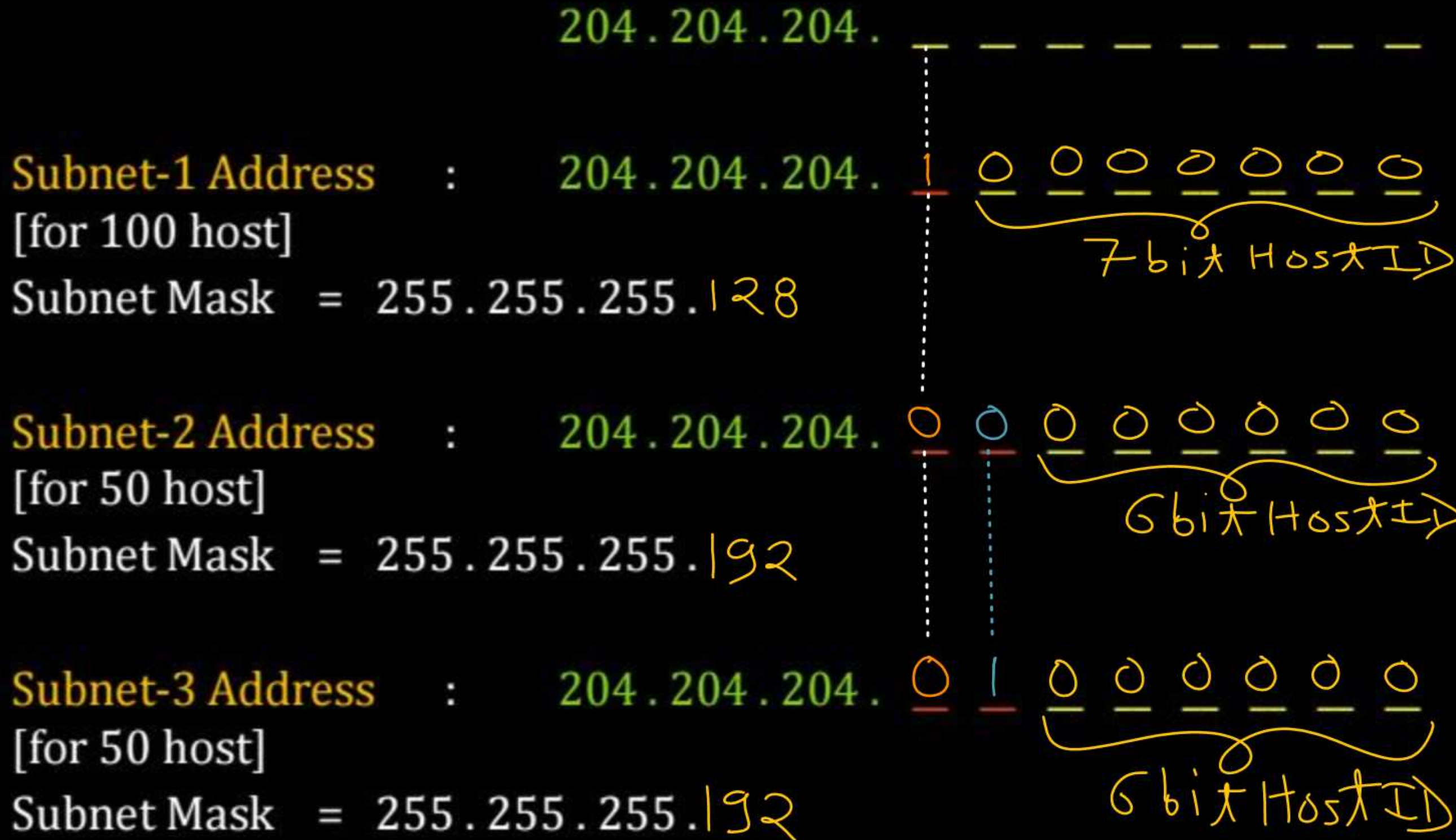
- ☒ (A) 204 . 204 . 204 . 128 / 255 . 255 . 255 . 192
 204 . 204 . 204 . 0 / 255 . 255 . 255 . 128
 204 . 204 . 204 . 64 / 255 . 255 . 255 . 128
- ☒ (B) 204 . 204 . 204 . 0 / 255 . 255 . 255 . 192
 204 . 204 . 204 . 192 / 255 . 255 . 255 . 128
 204 . 204 . 204 . 64 / 255 . 255 . 255 . 128
- ☒ (C) 204 . 204 . 204 . 128 / 255 . 255 . 255 . 128
 204 . 204 . 204 . 192 / 255 . 255 . 255 . 192
 204 . 204 . 204 . 224 / 255 . 255 . 255 . 192
- ☒ (D) 204 . 204 . 204 . 128 / 255 . 255 . 255 . 128
 204 . 204 . 204 . 64 / 255 . 255 . 255 . 192
 204 . 204 . 204 . 0 / 255 . 255 . 255 . 192

Ans: D

Class C network (Network Address : 204 . 204 . 204 . 0)



Class C network (Network Address : 204 . 204 . 204 . 0)



Subnet-1 Address	: 204.204.204.0	Subnet Mask	= 255.255.255.128
Subnet-2 Address	: 204.204.204.128	Subnet Mask	= 255.255.255.192
Subnet-3 Address	: 204.204.204.192	Subnet Mask	= 255.255.255.192

Subnet-1 Address	: 204.204.204.0	Subnet Mask	= 255.255.255.128
Subnet-2 Address	: 204.204.204.192	Subnet Mask	= 255.255.255.192
Subnet-3 Address	: 204.204.204.128	Subnet Mask	= 255.255.255.192

Subnet-1 Address	: 204 . 204 . 204 . 128	Subnet Mask	= 255 . 255 . 255 . 128
Subnet-2 Address	: 204 . 204 . 204 . 0	Subnet Mask	= 255 . 255 . 255 . 192
Subnet-3 Address	: 204 . 204 . 204 . 64	Subnet Mask	= 255 . 255 . 255 . 192



Subnet-1 Address	: 204 . 204 . 204 . 128	Subnet Mask	= 255 . 255 . 255 . 128
Subnet-2 Address	: 204 . 204 . 204 . 64	Subnet Mask	= 255 . 255 . 255 . 192
Subnet-3 Address	: 204 . 204 . 204 . 0	Subnet Mask	= 255 . 255 . 255 . 192

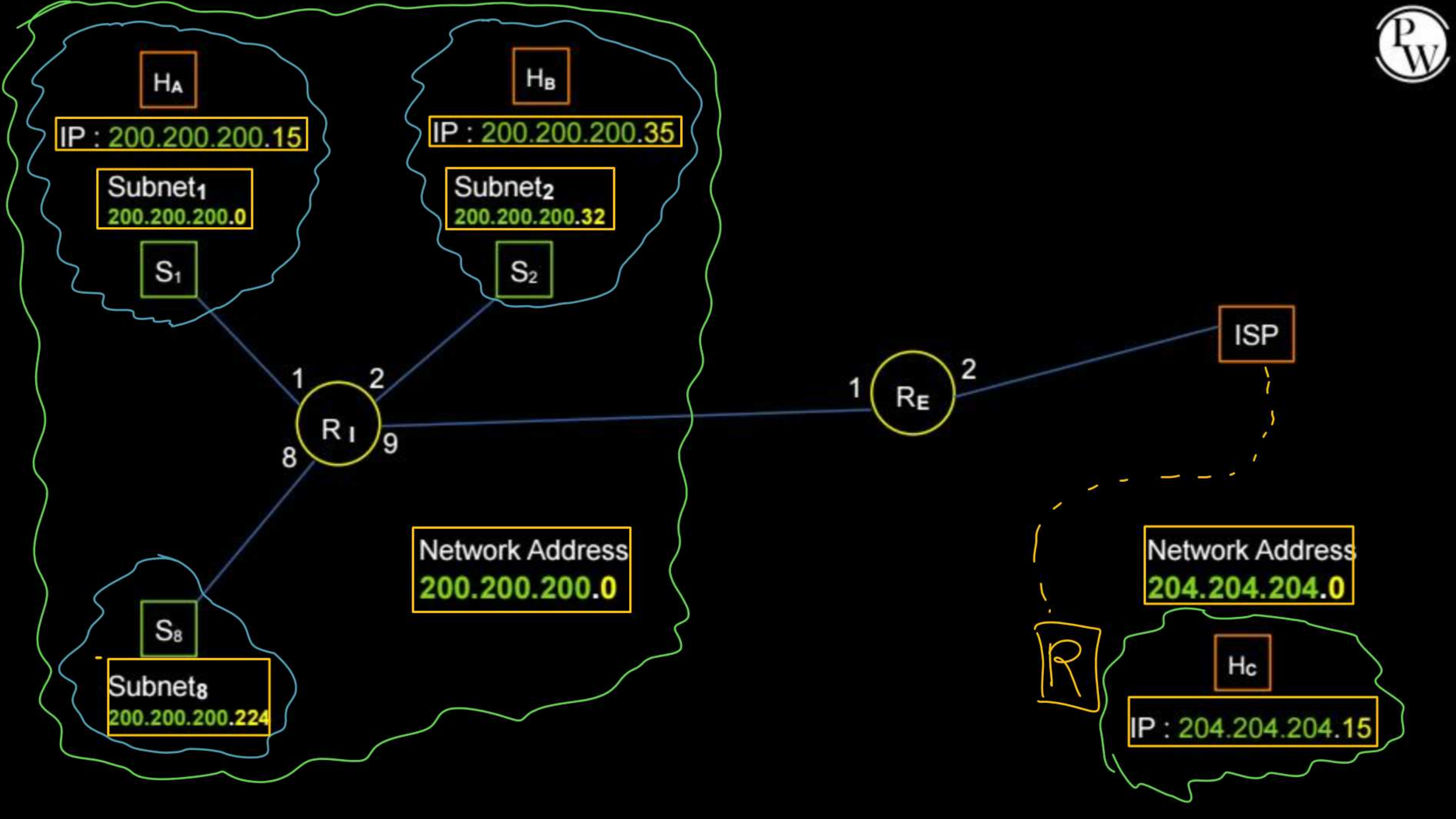
Class C Network :-

Network Address : 200 . 200 . 200 . 0
 Default Netmask : 255 . 255 . 255 . 0 $\Rightarrow$ 24 one's

After 3-bit subnetting :-

Subnet Mask : 255 . 255 . 255 . 224 $\Rightarrow$ 27 one's

1st Subnet Address : 200 . 200 . 200 . 0
 2nd Subnet Address : 200 . 200 . 200 . 32
 3rd Subnet Address : 200 . 200 . 200 . 64
 4th Subnet Address : 200 . 200 . 200 . 96
 5th Subnet Address : 200 . 200 . 200 . 128
 6th Subnet Address : 200 . 200 . 200 . 160
 7th Subnet Address : 200 . 200 . 200 . 192
 8th Subnet Address : 200 . 200 . 200 . 224





Topic : Forwarding Table

/ Routing Table



Router (**R_E**) forwarding table

Network Address	Network Mask	Interface ID	Next Hop
200 . 200 . 200 . 0	255 . 255 . 255 . 0	1	R _I
Default		2	ISP



Topic : Forwarding Table



Router (R_I) forwarding table

Subnet Address	Subnet Mask	Interface ID	Next Hop
200 . 200 . 200 . 0	255 . 255 . 255 . 224	1	On Link
200 . 200 . 200 . 32	255 . 255 . 255 . 224	2	On Link
200 . 200 . 200 . 64	255 . 255 . 255 . 224	3	On Link
⋮	⋮	⋮	
200 . 200 . 200 . 224	255 . 255 . 255 . 224	8	On Link
<u>Default</u>		9	R_E



Topic : Forwarding Table



CASE I :

Source IP Address : 200 . 200 . 200 . 15

Destination IP Address : 200 . 200 . 200 . 35

HA

HB

35 → 00100011
224 → 11100000
32 ← 00100000

Router	Interface ID	Next Hop
R _I	2	ON LINK

Destination IP Address : 200 . 200 . 200 . 35

Subnet Mask : 255 . 255 . 255 . 224

Subnet Address ← 200 . 200 . 200 . 32



Topic : Forwarding Table



CASE II :

Source IP Address : 200 . 200 . 200 . 35 H_B

Destination IP Address : 204 . 204 . 204 . 15 H_C

Router	Interface ID	Next Hop
R_I	9	R_E
R_E	2	ISP

R_I

IP Address : 204 . 204 . 204 . 15

Subnet Mask : 255 . 255 . 255 . 224

Subnet Add. $\leftarrow$ 204 . 204 . 204 .

R_E

IP Address : 204 . 204 . 204 . 15

Net Mask : 255 . 255 . 255 . 0

Net Add. $\leftarrow$ 204 . 204 . 204 . 0



Topic : Forwarding Table



CASE III:

Source IP Address : 204.204.204.15

HC

Destination IP Address : 200.200.200.35

HB

Router	Interface ID	Next Hop
[R _E]	1	R _I
[R _I]	2	ON LINK

R_E

R_I

IP Address : 200.200.200.35

~~Net Mask~~ : 255.255.255.0

Dest. Net Add. ← 200.200.200.0

IP Address : 200.200.200.35

~~Subnet Mask~~ : 255.255.255.224

Dest. Net Add. ← 200.200.200.32

Router's Routing Table

Network/ Subnetwork Address	Netmask/ Subnetmask	Next Hop
Add ₁	Mask ₁	R ₁
Add ₂	Mask ₂	R ₂

Dest. IP Add
(bit wise AND)

Mask₁

Result₁

if (Result₁ == Add₁)
→ Router forward packet
to R₁

Dest. IP Add
(bit wise AND)

Mask₂

Result₂

if (Result₂ == Add₂)
→ Router forward packet
to R₂



2 mins Summary



Topic

Routing Table

Topic

~~Stack~~



THANK - YOU

